MSG No. 68.520 C_Acca [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `m'm'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,2,3,4,29,30,31,32]`
2	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[5,6,7,8,25,26,27,28]`
3	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[9,10,11,12,21,22,23,24]`
4	$\left[0,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[13,14,15,16,17,18,19,20]`

Table 2: Wyckoff site: `4b`, site symmetry: `m'm'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[1,2,3,4,29,30,31,32]`
2	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[5,6,7,8,25,26,27,28]`
3	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[9,10,11,12,21,22,23,24]`
4	$\left[0,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[13,14,15,16,17,18,19,20]`

Table 3: Wyckoff site: `8c`, site symmetry: `2'/m'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,13,18,30]`
2	$[0,\ 0,\ 0]$	`[2,14,17,29]`
3	$\left[0,\ \frac{1}{2},\ 0\right]$	`[3,15,20,32]`
4	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,16,19,31]`
5	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[5,9,22,26]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[6,10,21,25]`
7	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[7,11,24,28]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[8,12,23,27]`

Table 4: Wyckoff site: `8d`, site symmetry: `.2'/m'.`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[1,5,27,31]`
2	$\left[\frac{1}{4},\ \frac{1}{4},\ 0\right]$	`[2,6,28,32]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ 0\right]$	`[3,7,25,29]`
4	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[4,8,26,30]`
5	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[9,13,19,23]`
6	$\left[\frac{3}{4},\ \frac{3}{4},\ 0\right]$	`[10,14,20,24]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ 0\right]$	`[11,15,17,21]`
8	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[12,16,18,22]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2/m'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,12,21,32]`
2	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[2,11,22,31]`
3	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[3,10,23,30]`
4	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[4,9,24,29]`
5	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[5,16,17,28]`
6	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[6,15,18,27]`
7	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[7,14,19,26]`
8	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[8,13,20,25]`

Table 6: Wyckoff site: `8f`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,12,18,27]`
2	$\left[\frac{1}{4}, 0, 0\right]$	`[2,11,17,28]`
3	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[3,10,20,25]`
4	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[4,9,19,26]`
5	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[5,16,22,31]`
6	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[6,15,21,32]`
7	$\left[\frac{3}{4}, 0, 0\right]$	`[7,14,24,29]`
8	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[8,13,23,30]`

Table 7: Wyckoff site: `8g`, site symmetry: `2m'm'`

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[1,2,31,32]`
2	$\left[-x, \frac{1}{4}, \frac{1}{4}\right]$	`[3,4,29,30]`
3	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	`[5,6,27,28]`
4	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[7,8,25,26]`
5	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[9,10,23,24]`
6	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	`[11,12,21,22]`
7	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	`[13,14,19,20]`
8	$\left[x, \frac{3}{4}, \frac{3}{4}\right]$	`[15,16,17,18]`

Table 8: Wyckoff site: `8h`, site symmetry: `m'2m'`

No.	position	mapping
1	$\left[0, y, \frac{1}{4}\right]$	`[1,3,30,32]`
2	$\left[0, \frac{1}{2}-y, \frac{1}{4}\right]$	`[2,4,29,31]`
3	$\left[\frac{1}{2}, \frac{1}{2}-y, \frac{3}{4}\right]$	`[5,7,26,28]`
4	$\left[\frac{1}{2}, y, \frac{3}{4}\right]$	`[6,8,25,27]`
5	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{4}\right]$	`[9,11,22,24]`
6	$\left[\frac{1}{2}, -y, \frac{1}{4}\right]$	`[10,12,21,23]`
7	$\left[0, -y, \frac{3}{4}\right]$	`[13,15,18,20]`

continued ...

Table 8

No.	position	mapping
8	$\left[0,\, y+\frac{1}{2},\, \frac{3}{4}\right]$	`[14,16,17,19]`

Table 9: Wyckoff site: `8i`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[0,\, \frac{1}{4},\, z\right]$	`[1,4,30,31]`
2	$\left[0,\, \frac{1}{4},\, \frac{1}{2}-z\right]$	`[2,3,29,32]`
3	$\left[\frac{1}{2},\, \frac{1}{4},\, -z\right]$	`[5,8,26,27]`
4	$\left[\frac{1}{2},\, \frac{1}{4},\, z+\frac{1}{2}\right]$	`[6,7,25,28]`
5	$\left[\frac{1}{2},\, \frac{3}{4},\, z\right]$	`[9,12,22,23]`
6	$\left[\frac{1}{2},\, \frac{3}{4},\, \frac{1}{2}-z\right]$	`[10,11,21,24]`
7	$\left[0,\, \frac{3}{4},\, -z\right]$	`[13,16,18,19]`
8	$\left[0,\, \frac{3}{4},\, z+\frac{1}{2}\right]$	`[14,15,17,20]`

Table 10: Wyckoff site: `16j`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{1}{4},\, \frac{1}{2},\, z\right]$	`[1,12]`
2	$\left[\frac{1}{4},\, 0,\, \frac{1}{2}-z\right]$	`[2,11]`
3	$\left[\frac{3}{4},\, \frac{1}{2},\, \frac{1}{2}-z\right]$	`[3,10]`
4	$\left[\frac{3}{4},\, 0,\, z\right]$	`[4,9]`
5	$\left[\frac{1}{4},\, 0,\, -z\right]$	`[5,16]`
6	$\left[\frac{1}{4},\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[6,15]`
7	$\left[\frac{3}{4},\, 0,\, z+\frac{1}{2}\right]$	`[7,14]`
8	$\left[\frac{3}{4},\, \frac{1}{2},\, -z\right]$	`[8,13]`
9	$\left[\frac{1}{4},\, 0,\, z+\frac{1}{2}\right]$	`[17,28]`
10	$\left[\frac{1}{4},\, \frac{1}{2},\, -z\right]$	`[18,27]`
11	$\left[\frac{3}{4},\, 0,\, -z\right]$	`[19,26]`
12	$\left[\frac{3}{4},\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[20,25]`
13	$\left[\frac{1}{4},\, \frac{1}{2},\, \frac{1}{2}-z\right]$	`[21,32]`
14	$\left[\frac{1}{4},\, 0,\, z\right]$	`[22,31]`
15	$\left[\frac{3}{4},\, \frac{1}{2},\, z\right]$	`[23,30]`
16	$\left[\frac{3}{4},\, 0,\, \frac{1}{2}-z\right]$	`[24,29]`

Table 11: Wyckoff site: `16k`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, \frac{1}{2}\right]$	`[1,27]`
2	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, 0\right]$	`[2,28]`
3	$\left[\frac{3}{4},\, y,\, 0\right]$	`[3,25]`
4	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	`[4,26]`
5	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	`[5,31]`
6	$\left[\frac{1}{4},\, y,\, 0\right]$	`[6,32]`

continued ...

Table 11

No.	position	mapping
7	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ 0\right]$	[7,29]
8	$\left[\frac{3}{4},\ y,\ \frac{1}{2}\right]$	[8,30]
9	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[9,19]
10	$\left[\frac{3}{4},\ -y,\ 0\right]$	[10,20]
11	$\left[\frac{1}{4},\ y+\frac{1}{2},\ 0\right]$	[11,17]
12	$\left[\frac{1}{4},\ -y,\ \frac{1}{2}\right]$	[12,18]
13	$\left[\frac{3}{4},\ -y,\ \frac{1}{2}\right]$	[13,23]
14	$\left[\frac{3}{4},\ y+\frac{1}{2},\ 0\right]$	[14,24]
15	$\left[\frac{1}{4},\ -y,\ 0\right]$	[15,21]
16	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[16,22]

Table 12: Wyckoff site: 16l, site symmetry: 2'..

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[1,18]
2	$[x,\ 0,\ 0]$	[2,17]
3	$\left[-x,\ \frac{1}{2},\ 0\right]$	[3,20]
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[4,19]
5	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[5,22]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[6,21]
7	$\left[x+\frac{1}{2},\ 0,\ 0\right]$	[7,24]
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[8,23]
9	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[9,26]
10	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[10,25]
11	$\left[\frac{1}{2}-x,\ 0,\ 0\right]$	[11,28]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[12,27]
13	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[13,30]
14	$[-x,\ 0,\ 0]$	[14,29]
15	$\left[x,\ \frac{1}{2},\ 0\right]$	[15,32]
16	$\left[x,\ 0,\ \frac{1}{2}\right]$	[16,31]

Table 13: Wyckoff site: 16m, site symmetry: m'..

No.	position	mapping
1	$[0,\ y,\ z]$	[1,30]
2	$\left[0,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[2,29]
3	$\left[0,\ y,\ \frac{1}{2}-z\right]$	[3,32]
4	$\left[0,\ \frac{1}{2}-y,\ z\right]$	[4,31]
5	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[5,26]
6	$\left[\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[6,25]
7	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[7,28]
8	$\left[\frac{1}{2},\ y,\ -z\right]$	[8,27]
9	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[9,22]
10	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[10,21]

continued ...

Table 13

No.	position	mapping
11	$\left[\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[11,24]
12	$\left[\frac{1}{2},\, -y,\, z\right]$	[12,23]
13	$[0,\, -y,\, -z]$	[13,18]
14	$\left[0,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[14,17]
15	$\left[0,\, -y,\, z+\frac{1}{2}\right]$	[15,20]
16	$\left[0,\, y+\frac{1}{2},\, -z\right]$	[16,19]

Table 14: Wyckoff site: 16n, site symmetry: .m'.

No.	position	mapping
1	$\left[x,\, \frac{1}{4},\, z\right]$	[1,31]
2	$\left[x,\, \frac{1}{4},\, \frac{1}{2}-z\right]$	[2,32]
3	$\left[-x,\, \frac{1}{4},\, \frac{1}{2}-z\right]$	[3,29]
4	$\left[-x,\, \frac{1}{4},\, z\right]$	[4,30]
5	$\left[\frac{1}{2}-x,\, \frac{1}{4},\, -z\right]$	[5,27]
6	$\left[\frac{1}{2}-x,\, \frac{1}{4},\, z+\frac{1}{2}\right]$	[6,28]
7	$\left[x+\frac{1}{2},\, \frac{1}{4},\, z+\frac{1}{2}\right]$	[7,25]
8	$\left[x+\frac{1}{2},\, \frac{1}{4},\, -z\right]$	[8,26]
9	$\left[x+\frac{1}{2},\, \frac{3}{4},\, z\right]$	[9,23]
10	$\left[x+\frac{1}{2},\, \frac{3}{4},\, \frac{1}{2}-z\right]$	[10,24]
11	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, \frac{1}{2}-z\right]$	[11,21]
12	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, z\right]$	[12,22]
13	$\left[-x,\, \frac{3}{4},\, -z\right]$	[13,19]
14	$\left[-x,\, \frac{3}{4},\, z+\frac{1}{2}\right]$	[14,20]
15	$\left[x,\, \frac{3}{4},\, z+\frac{1}{2}\right]$	[15,17]
16	$\left[x,\, \frac{3}{4},\, -z\right]$	[16,18]

Table 15: Wyckoff site: 16o, site symmetry: ..m'

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{4}\right]$	[1,32]
2	$\left[x,\, \frac{1}{2}-y,\, \frac{1}{4}\right]$	[2,31]
3	$\left[-x,\, y,\, \frac{1}{4}\right]$	[3,30]
4	$\left[-x,\, \frac{1}{2}-y,\, \frac{1}{4}\right]$	[4,29]
5	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	[5,28]
6	$\left[\frac{1}{2}-x,\, y,\, \frac{3}{4}\right]$	[6,27]
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	[7,26]
8	$\left[x+\frac{1}{2},\, y,\, \frac{3}{4}\right]$	[8,25]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{4}\right]$	[9,24]
10	$\left[x+\frac{1}{2},\, -y,\, \frac{1}{4}\right]$	[10,23]
11	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{4}\right]$	[11,22]
12	$\left[\frac{1}{2}-x,\, -y,\, \frac{1}{4}\right]$	[12,21]
13	$\left[-x,\, -y,\, \frac{3}{4}\right]$	[13,20]
14	$\left[-x,\, y+\frac{1}{2},\, \frac{3}{4}\right]$	[14,19]

continued ...

Table 15

No.	position	mapping
15	$\left[x,\, -y,\, \frac{3}{4}\right]$	[15,18]
16	$\left[x,\, y+\frac{1}{2},\, \frac{3}{4}\right]$	[16,17]

Table 16: Wyckoff site: 32p, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[2]
3	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[3]
4	$\left[-x,\, \frac{1}{2}-y,\, z\right]$	[4]
5	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[5]
6	$\left[\frac{1}{2}-x,\, y,\, z+\frac{1}{2}\right]$	[6]
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[7]
8	$\left[x+\frac{1}{2},\, y,\, -z\right]$	[8]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[9]
10	$\left[x+\frac{1}{2},\, -y,\, \frac{1}{2}-z\right]$	[10]
11	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\, -y,\, z\right]$	[12]
13	$[-x,\, -y,\, -z]$	[13]
14	$\left[-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[14]
15	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[15]
16	$\left[x,\, y+\frac{1}{2},\, -z\right]$	[16]
17	$\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[17]
18	$[x,\, -y,\, -z]$	[18]
19	$\left[-x,\, y+\frac{1}{2},\, -z\right]$	[19]
20	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[20]
21	$\left[\frac{1}{2}-x,\, -y,\, \frac{1}{2}-z\right]$	[21]
22	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[22]
23	$\left[x+\frac{1}{2},\, -y,\, z\right]$	[23]
24	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[24]
25	$\left[x+\frac{1}{2},\, y,\, z+\frac{1}{2}\right]$	[25]
26	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[26]
27	$\left[\frac{1}{2}-x,\, y,\, -z\right]$	[27]
28	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[28]
29	$\left[-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[29]
30	$[-x,\, y,\, z]$	[30]
31	$\left[x,\, \frac{1}{2}-y,\, z\right]$	[31]
32	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[32]